

dbt Training

● Beginner · Module 06 · 90 min

Testing

Data Quality

Not optional. CI rejects Silver and Gold PRs that are missing mandatory tests on key columns.

Why Tests Are Mandatory

****Without tests, transformations break silently**** - A column renamed in Bronze → staging model silently returns NULLs - Duplicate surrogate keys → Power BI relationships corrupt - NULL in a required column → propagates downstream undetected - A Gold mart shows wrong numbers → business discovers it weeks later
The failure is always silent. The discovery is always late. The cost is always high.

****Tests are a CI requirement****

Silver and Gold PRs with missing tests on `_key` columns → **rejected**

Missing FK tests on fact tables → **rejected**

Zero tests on a new Silver model → **rejected**

Reference: `dbt-test-strategy` skill — full placement guide per layer

Two Types of dbt Tests

****Generic tests — 95% of what you'll write**** Defined in ``schema.yml``. Reusable. Parameterised.

```
models:
  - name: fct_prescription
    columns:
      - name: prescription_key
        tests:
          - unique
          - not_null
      - name: patient_key
        tests:
          - relationships:
              to: ref('dim_patient')
              field: patient_key
```

****Singular tests — for complex business rules**** Standalone `.sql` files in ``tests/``. Return rows on failure. Two common patterns: ****Pattern 1 — business rule on aggregates (``GROUP BY ... HAVING``)****

```
-- tests/assert_fct_revenue_no_negative_amounts.sql
SELECT customer_key, SUM(amount_net) AS total
FROM {{ ref('fct_revenue') }}
GROUP BY 1
HAVING total < 0
```

****Pattern 2 — orphan FK check (`LEFT JOIN ... WHERE IS NULL`):****

```
-- tests/assert_no_orphan_prescriptions.sql
SELECT p.prescription_key
FROM {{ ref('fct_prescription') }} p
LEFT JOIN {{ ref('dim_patient') }} d ON p.patient_key = d.patient_key
WHERE d.patient_key IS NULL
```

Use singular tests only when the rule can't be expressed in YAML. Harder to read — prefer generic.

The Four Generic Tests

unique

```
- name: prescription_key
  tests:
    - unique
```

Fails if any value appears more than once. Required on all `_key` columns.

not_null

```
- name: prescription_key
  tests:
    - not_null
```

Fails if any value is NULL. Required on all `_key` columns.

accepted_values

```
- name: medication_type
  tests:
    - accepted_values:
        values: ['tablet', 'liquid', 'injection']
```

Fails if any value outside the list appears. Use for status/type columns.

relationships

```
- name: patient_key
  tests:
    - relationships:
        to: ref('dim_patient')
        field: patient_key
```

FK integrity check. Required on all FK columns in Silver facts and Gold marts.

Test Severity and `dbt build`

****Severity: `error` vs `warn`**** Global default — `dbt\_project.yml`:

```
tests:
  analytics:
    +severity: warn    # project-wide default
  silver:
    +severity: error # Silver always errors
  gold:
    +severity: error # Gold always errors
```

Per individual test — `schema.yml`:

```
- name: prescription_key
  tests:
    - unique:
        config:
          severity: error # ← CI stops, pipeline halts
- name: dosage_amount
  tests:
    - not_null:
        config:
          severity: warn  # ← logged, continues
```

Per-test overrides the global setting.

****`dbt build` — the only correct command****

```
# ❌ WRONG
dbt run && dbt test

# ✅ CORRECT — always
dbt build
```

`dbt build` runs models and tests **in DAG order**. If `dim\_patient` fails a test, `fct\_prescription` (which depends on it) is **never built**. `dbt run` && `dbt test` runs all models first — loading bad data into `fct\_prescription` before the tests even run.

Never use `dbt run && dbt test` in CI. Always `dbt build`.

Mandatory Test Requirements

| Column type | Required tests | |---|---| | `\_key` columns (surrogate PKs) | `unique` + `not\_null` (both, always) | | FK columns (referencing another model) | `relationships` | | Status / type columns | `accepted\_values` (if finite value set exists) | | Business-critical measures | `not\_null` |

CI check: Silver and Gold models must have at least one test per `_key` column. A PR with zero tests on a new Silver model will not pass review. This is checked manually via the `dbt-test-strategy` skill until CI automation is complete. Full test placement guide (when to test in Bronze vs Silver vs Gold, store_failures config, dbt-expectations) → `dbt-test-strategy` skill

Exercise: Write Tests for `fct_prescription` (30 min)

The model has no tests. Add the complete test suite to `schema.yml`.

| Column | Type | Notes | |---|---|---| | `prescription\_key` | Surrogate PK | MD5 hash | | `patient\_key` | FK → `dim\_patient` | — | | `doctor\_key` | FK → `dim\_doctor` | — | | `prescription\_date` | Date | Required | | `medication\_type` | String | `tablet`, `liquid`, `injection`, `topical` | | `dosage\_amount` | Number | Can be null — not yet confirmed | | `notes` | String | Optional free text |

Step 1

Write full schema.yml block with all required tests

Step 2

Add warn-severity not_null test on dosage_amount

Step 3

Introduce a duplicate prescription_key in dev. Run dbt test. Document the failure output.

layout: center

Module 06 Complete

Next: Module 07

Documentation — the final Beginner module

Prep Q1: Generic test vs singular test — difference?

Prep Q2: Name the four built-in generic tests

Prep Q3: What does dbt build do differently from dbt run && dbt test?

Prep Q4: Two mandatory tests on every `_key` column in Silver?